

Newman-Penrose Formalism Calculations

Stephani et al., Equation (12.8)

Metric

$$ds^2 = -B^2 \Sigma (z, k')^2 dt \otimes dt + A^2 dx \otimes dx + A^2 \Sigma (x, k)^2 dy \otimes dy + B^2 dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} B \Sigma (z, k') \right) dt + \left(\frac{1}{2} \sqrt{2} B \right) dz$$

$$n = \left(\frac{1}{2} \sqrt{2} B \Sigma (z, k') \right) dt + \left(-\frac{1}{2} \sqrt{2} B \right) dz$$

$$m = \left(\frac{1}{2} \sqrt{2} A \right) dx + \left(\frac{1}{2} i \sqrt{2} A \Sigma (x, k) \right) dy$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} A \right) dx + \left(-\frac{1}{2} i \sqrt{2} A \Sigma (x, k) \right) dy$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2 B \Sigma (z, k')} \right) dt + \left(\frac{\sqrt{2}}{2 B} \right) dz$$

$$n = \left(-\frac{\sqrt{2}}{2B\Sigma(z, k')} \right) dt + \left(-\frac{\sqrt{2}}{2B} \right) dz$$

$$m = \left(\frac{\sqrt{2}}{2A} \right) dx + \left(\frac{i\sqrt{2}}{2A\Sigma(x, k)} \right) dy$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2A} \right) dx + \left(-\frac{i\sqrt{2}}{2A\Sigma(x, k)} \right) dy$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2B\Sigma(z, k')} \partial_t + \frac{\sqrt{2}}{2B} \partial_z$$

$$\Delta = -\frac{\sqrt{2}}{2B\Sigma(z, k')} \partial_t - \frac{\sqrt{2}}{2B} \partial_z$$

$$\delta = \frac{\sqrt{2}}{2A} \partial_x + \frac{i\sqrt{2}}{2A\Sigma(x, k)} \partial_y$$

$$\bar{\delta} = \frac{\sqrt{2}}{2A} \partial_x - \frac{i\sqrt{2}}{2A\Sigma(x, k)} \partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\frac{\partial}{\partial x}\Sigma(x,k)}{4A\Sigma(x,k)}$$

$$\beta = \frac{\sqrt{2}\frac{\partial}{\partial x}\Sigma(x,k)}{4A\Sigma(x,k)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{\sqrt{2}\frac{\partial}{\partial z}\Sigma(z,k')}{4B\Sigma(z,k')}$$

$$\gamma = \frac{\sqrt{2}\frac{\partial}{\partial z}\Sigma(z,k')}{4B\Sigma(z,k')}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{B^2 \Sigma \left(z, k' \right) \frac{\partial^2}{(\partial x)^2} \Sigma (x, k) - A^2 \Sigma (x, k) \frac{\partial^2}{(\partial z)^2} \Sigma \left(z, k' \right)}{4 A^2 B^2 \Sigma (x, k) \Sigma (z, k')}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = -\frac{B^2 \Sigma \left(z, k' \right) \frac{\partial^2}{(\partial x)^2} \Sigma (x, k) + A^2 \Sigma (x, k) \frac{\partial^2}{(\partial z)^2} \Sigma \left(z, k' \right)}{12 A^2 B^2 \Sigma (x, k) \Sigma (z, k')}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{B^2 \Sigma \left(z, k' \right) \frac{\partial^2}{(\partial x)^2} \Sigma (x, k) + A^2 \Sigma (x, k) \frac{\partial^2}{(\partial z)^2} \Sigma \left(z, k' \right)}{6 A^2 B^2 \Sigma (x, k) \Sigma (z, k')}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "*D*"